Saathvik Kalepu

Software Engineer | Backend & Distributed Systems | Applied AI

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Education

Indian Institute of Information Technology, Design & Manufacturing (IIITDM) Kurnool, AP, India

B.Tech in Computer Science & Engineering — ***CGPA: 8.2 / 10.0*** *Aug. 2023 – May 2027*

- Coursework: Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks, Distributed Systems
- Class XII: **98.8%** (Valley Oak Junior College, 2023) | Class X: **10.0 / 10.0 GPA** (2021)

Experience

AI Engineer Intern Aug. 2026 – Present

Darwix AI *Gurugram, India*

- Built an end-to-end **AI hiring automation platform** (**Python, Flask, MySQL, OpenAI**) that runs the full 9-stage recruiting funnel: WhatsApp outreach, AI voice screening, assessments, shortlisting, interviews, and offers.
- Developed **LLM-powered voice screening agents** (Vapi, GPT-4o-mini) that call every candidate, score fit from call transcripts, auto-reject poor fits, and route qualified candidates to intern or full-time tracks.
- Made it config-driven with **12 role-specific screening profiles** so any team can reuse it; processed **1,500+** candidates across **80+** campaigns, with assessments sent within **2 minutes** via webhooks.
- Built a real-time AI video verification agent for insurance pre-issuance checks (**FastAPI**, Deepgram STT, ElevenLabs TTS) with a deterministic dialogue state machine, backed by **226 automated tests**.

Software Development Intern Apr. 2026 – Jul. 2026

Caterworld.ai *Chennai, India*

- Built and maintained **CI/CD pipelines** automating build, test, and deployment across **10 application modules** of the Caterworld platform, replacing manual releases.
- Developed **Progressive Web App (PWA)** modules for the main website in **React.js** (service workers, offline caching, installable app shell), improving frontend load performance by **15%**.

Software Engineering Intern Jul. 2024 – Jan. 2025

AIMaster.live *Hyderabad, India*

- Engineered **20+ RESTful backend API endpoints** in **Python/Flask** and **Node.js** covering authentication, session management, and request workflows.
- Optimized SQL queries to cut retrieval latency by **15%**; shipped user-facing features in **TypeScript/React**.

Projects

QuorumDB — Distributed Key-Value Store & Consensus Engine | *Go, Raft, LSM Tree, Docker* GitHub

- Built a replicated key-value store in **Go** with zero third-party dependencies: **Raft** leader election and log replication, Dynamo-style quorum reads/writes, consistent hashing, and hinted handoff.
- Built an **LSM-tree** storage engine: WAL, memtable, bloom filters, sparse index, compaction, crash recovery.
- Chaos-tested by killing random nodes under concurrent load: **17,658 writes, 0 failures**; leader failover in **∼2s**.

ClaimFlow — Healthcare Claims & Prior-Authorization Platform | *FastAPI, PostgreSQL, Redis, scikit-learn* 2026

- Built a claims-processing backend with a **gradient-boosting denial-risk model** (15 features, per-claim risk explanations) and **GPT-4o-mini** extraction of ICD-10/CPT codes from clinical notes, with a regex fallback.
- Added JWT auth with **RBAC**, an append-only audit log, **Redis** TTL caching for eligibility checks, and Prometheus metrics; containerized with Docker Compose and covered by **20** pytest tests.

Switchboard — Intent-Classified Banking Agent | *Python, HuggingFace Transformers, LangChain* GitHub

- Trained a 77-intent classifier on **13K** real banking queries: **93% accuracy**, 98.4% top-3 accuracy on 3,080 held-out examples; its prediction guides a LangChain tool-calling agent.
- Built a reproducible agent eval suite; fixing a tool-selection failure it caught raised pass rate from **67% to 83%**.

Technical Skills

Languages: Python, Java, C/C++, JavaScript, TypeScript, SQL, Go
Frameworks & Web: React.js, Node.js, Express, Spring Boot, FastAPI, Flask, REST APIs, Microservices, PWA, HTML5, CSS3
AI & ML: LLMs, RAG, LangChain, HuggingFace Transformers, Voice AI Agents, LLM Evaluation, scikit-learn, pandas
Databases & Tools: PostgreSQL, MySQL, MongoDB, Redis, SQLite, Docker, Git, AWS, Linux, Prometheus, CI/CD

Certifications & Achievements

Problem Solving: **350+ LeetCode problems** solved (35 Hard, 150+ Medium) | Member, Google Developer Groups on Campus
Certifications: NPTEL Elite Cloud Computing, IIT Kharagpur | Microsoft Azure AI Essentials | Cisco CCST Networking